

LETHBRIDGE CDA, AB, Canada

WMO: 712430

Lat: 49.700N

Lon: 112.770W

Elev: 910

StdP: 90.86

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-24.4	-31.5	0.2	-27.6	-27.9	0.3	-24.3	15.9	4.7	14.4	5.7	2.4	200	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	31.8	17.0	29.8	16.4	27.8	16.0	19.1	27.4	18.0	26.4	16.9	25.3	4.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.9	22.5	14.9	11.8	21.3	13.7	10.9	20.0	58.1	27.4	54.3	26.6	51.0	25.3	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.8	10.5	-32.1	34.7	3.7	1.6	-34.8	35.9	-37.0	36.8	-39.0	37.8	-41.7	39.0		
(5)			-32.3	21.4	3.6	1.4	-34.9	22.5	-37.1	23.3	-39.1	24.1	-41.8	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-5.5	-4.2	0.0	5.9	11.4	15.5	19.1	18.4	13.6	7.0	0.2	-4.8	(6)
(7)		DBStd	10.84	9.60	8.55	8.00	5.09	4.54	3.35	3.04	3.38	4.67	5.36	7.57	8.47	(7)
(8)		HDD10.0	2261	481	397	313	137	37	2	0	1	19	119	297	457	(8)
(9)		HDD18.3	4456	739	630	569	372	218	96	26	42	152	351	544	716	(9)
(10)		CDD10.0	967	1	1	2	15	82	167	284	260	127	27	3	0	(10)
(11)		CDD18.3	121	0	0	0	0	3	12	52	43	10	1	0	0	(11)
(12)		CDH23.3	1957	0	0	1	8	98	180	737	696	232	7	0	0	(12)
(13)		CDH26.7	662	0	0	0	1	21	42	263	261	74	1	0	0	(13)
(14)	Wind	WSAvg	4.5	5.0	4.5	4.7	4.8	4.5	4.3	3.8	3.5	3.9	4.7	4.9	5.0	(14)
(15)	Precipitation	PrecAvg	425	24	16	28	40	54	67	45	46	18	18	23	(15)	
(16)		PrecMax	766	59	37	78	169	134	169	123	143	140	46	51	58	(16)
(17)		PrecMin	243	5	2	5	0	10	3	7	4	4	0	0	4	(17)
(18)		PrecStd	111	13	8	16	33	30	42	32	32	35	13	14	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	14.1	18.3	24.4	29.1	31.0	34.4	34.1	32.0	24.2	18.0	12.5	(19)
(20)			MCWB	6.3	5.7	8.2	11.4	14.5	17.4	18.1	16.8	15.3	11.9	8.9	5.5	(20)
(21)		2%	DB	10.3	11.2	15.3	20.7	26.2	27.5	31.7	31.8	28.8	20.8	14.5	9.8	(21)
(22)			MCWB	4.6	4.7	7.1	9.7	13.6	15.9	18.1	16.4	14.5	11.0	7.3	4.2	(22)
(23)		5%	DB	8.3	8.9	12.9	17.6	23.3	25.3	29.6	29.6	26.0	18.0	12.0	7.9	(23)
(24)			MCWB	3.5	3.2	5.8	8.2	12.2	15.2	17.6	15.9	13.9	9.8	6.0	3.0	(24)
(25)		10%	DB	6.6	6.8	10.4	14.9	20.6	23.2	27.5	27.5	22.9	15.4	9.8	6.1	(25)
(26)			MCWB	2.4	1.9	4.4	7.0	11.0	14.1	17.0	15.6	12.7	8.2	4.5	1.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	6.6	9.1	12.0	15.5	19.3	21.6	19.1	16.8	13.1	10.1	6.3	(27)
(28)			MCD B	12.4	12.9	17.0	22.8	26.2	27.2	29.5	28.9	28.0	22.1	16.8	11.2	(28)
(29)		2%	WB	4.9	4.9	7.5	10.2	14.0	17.4	19.8	17.9	15.4	11.4	7.8	4.6	(29)
(30)			MCD B	9.7	10.4	14.3	19.2	24.5	25.1	27.8	27.5	26.3	19.6	13.7	9.3	(30)
(31)		5%	WB	3.7	3.5	6.0	8.7	12.8	16.0	18.7	17.0	14.2	10.0	6.2	3.2	(31)
(32)			MCD B	8.0	8.6	12.2	16.8	21.9	23.3	26.7	26.2	24.4	17.4	11.6	7.4	(32)
(33)		10%	WB	2.5	2.1	4.6	7.3	11.6	14.8	17.7	16.2	13.1	8.6	4.8	1.9	(33)
(34)			MCD B	6.5	6.6	10.1	14.3	19.4	21.6	25.8	25.6	22.1	14.9	9.6	6.1	(34)
(35)	Mean Daily Temperature Range	MDBR	10.7	11.2	11.5	12.2	12.6	11.8	14.2	15.0	13.7	11.9	10.5	10.3	(35)	
(36)		5% DB	MCD BR	11.0	13.0	14.9	17.4	17.5	15.9	18.1	19.6	19.7	15.9	12.9	10.8	(36)
(37)			MCWBR	7.3	8.0	8.5	8.8	7.8	7.0	7.7	7.5	8.1	8.2	7.7	7.0	(37)
(38)		5% WB	MCD BR	11.1	13.1	14.7	16.3	16.2	14.1	15.7	16.7	18.0	15.4	12.8	11.0	(38)
(39)			MCWBR	7.6	8.4	8.7	8.5	7.4	6.9	7.7	7.1	7.7	8.1	7.9	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.259	0.283	0.319	0.332	0.330	0.333	0.350	0.312	0.280	0.256	0.234	(40)	
(41)		taud	2.458	2.491	2.484	2.392	2.407	2.461	2.468	2.396	2.474	2.542	2.505	2.458	(41)	
(42)		Ebn at Noon	831	912	938	928	926	928	919	885	891	872	818	805	(42)	
(43)		Edh at Noon	58	73	88	108	111	106	104	107	88	69	55	50	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.24	2.24	3.50	4.84	5.77	6.42	7.05	5.60	3.98	2.48	1.38	0.98	(44)	
(45)		RadStd	0.10	0.18	0.29	0.33	0.48	0.46	0.36	0.34	0.33	0.25	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page